

Reducing Repeated LLM Calls Through Caching

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BACKGROUND & RESEARCH QUESTIONS

- LLM called for every request — even exact duplicates
- Paraphrased, reworded queries dominate real usage
- Caching is proven in web engineering, underused in AI
- Redundant inference has a real latency, cost and compute footprint

PROBLEM OVERVIEW

LLM-backed FAQ systems routinely re-answer questions that only differ in wording — paying full inference cost and latency for a query the system has, in effect, already seen. This project puts a two-stage cache (exact match, then FAISS semantic search) in front of a local Llama 3 model and measures how much redundant inference it removes.

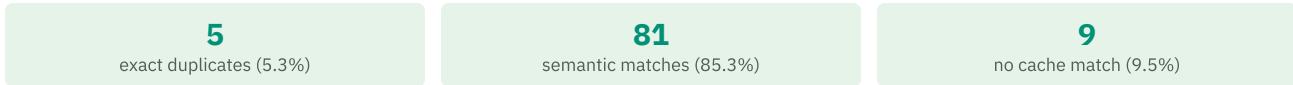
RESEARCH QUESTIONS

- What proportion of requests can be served from cache instead of the LLM?
- How much do caching strategies reduce model calls, latency, and cost?
- How does semantic caching compare with exact-match for answer quality and safe reuse?

METHODOLOGY

DATASET & QUERY SET

95 test queries, paraphrased from 115 original university FAQ entries



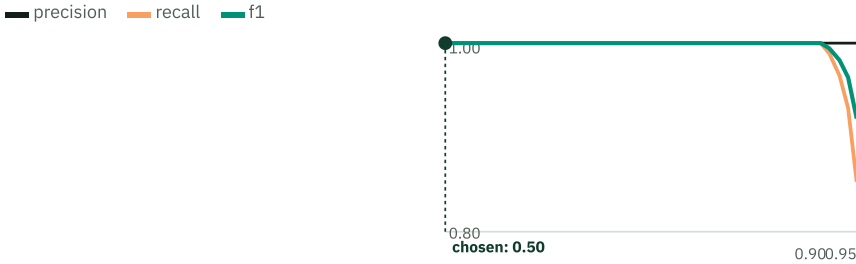
THREE SYSTEM VARIANTS

No-Cache Baseline	95 calls · 5.89s avg
Exact-Match Cache	90 calls · 5.14s avg
Semantic Cache (FAISS)	9 calls · 0.54s avg

SEMANTIC MATCHING PIPELINE

- Query received, normalized
- Exact-match cache checked first (normalized string compare)
- Miss → encoded with all-MiniLM-L6-v2 into a 384-dim vector
- FAISS flat index returns nearest neighbor + cosine score
- Score ≥ 0.50 → reuse cached answer; else call Llama 3 & cache it

Threshold Sweep — F1, Precision & Recall



BORDERLINE FAILURE CASE

1 of 81 semantic hits (1.2%) is an **incorrect reuse** despite clearing the 0.50 similarity gate — string similarity and topic overlap both stayed below the safe-reuse rubric. Safety verdict: consider raising the threshold for production use.

REPRODUCIBILITY

Python 3.9+ Ollama · Llama 3 FAISS Sentence-Transformers

```
$ pip install -r requirements.txt
$ ollama run llama3
$ python main.py
```

Benchmark: `python evaluation/benchmark.py --shuffle --seed 42`. Set `MOCK_LLM=1` to reproduce the results above without a live Ollama server.

All five research questions, threshold tuning, and the borderline-safety audit are reproducible from the evaluation suite — `benchmark.py`, `threshold_tests.py`, `multi_run.py`, `borderline_inspection.py`, `rq_analysis.py` — against the same 95-query test set.

MAIN RESULTS

91%

fewer LLM calls

11×

faster latency

90.5%

cache hit rate

Metric	No Cache	Exact Cache	Semantic Cache
Cache hit rate	0%	5.3%	90.5%
LLM calls (of 95)	95	90	9
Avg latency	5.89 s	5.14 s	0.54 s
Latency 95% CI	[5.31, 6.47]	[4.66, 5.61]	[0.22, 0.87]
Est. cost / query	\$0.0000465	\$0.0000448	\$0.0000035
Cost reduction vs. no-cache	—	2%	91%
Total tokens (test set)	2,209	2,129	168
Answer exact-match vs. gold	0%	5.3%	89.5%
String similarity vs. gold	0.30	0.34	0.93

TRADE-OFF ANALYSIS — ALL RESEARCH QUESTIONS



DISCUSSION & CONCLUSION

Why the hit rate is so high (90.5%)

FAQ questions cluster into a small set of underlying intents; template paraphrasing stays semantically close to the cached originals, so a 0.50 cosine threshold catches nearly all of them.

When do we need semantic caching?

Exact-match alone is close to useless once users reword questions (5.3% hit rate); semantic caching is what actually absorbs paraphrase traffic in a live FAQ system.

Semantic similarity ≠ semantic correctness

1 of 81 semantic hits (1.2%) reused a factually wrong answer despite clearing the similarity gate — embedding distance is a proxy for meaning, not a guarantee of it.

Limitations

Template paraphrases, a small 115-entry FAQ set, one embedding model, and local inference — results may shift with real user phrasing or a larger corpus.

CONCLUSION — KEY RESEARCH FINDINGS

- RQ1** 90.5% of paraphrased queries are served from cache without touching the LLM.
- RQ2** Caching cuts LLM calls **91%**, latency **11×**, and estimated cost **91%** vs. calling the model every time.
- RQ3** Semantic-cache answers match gold **89.5%** of the time — far above exact-match's 5.3% — with one flagged unsafe reuse in 81 hits.

REFERENCES

- [1] all-MiniLM-L6-v2 sentence embeddings — sentence-transformers
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- [3] Ollama — local Llama 3 (8B, Q4_0) inference
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- [5] Reimers & Gurevych (2019) — Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks
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